# Introduction to EDA
# (Exploratory Data Analysis)

IRON HACK

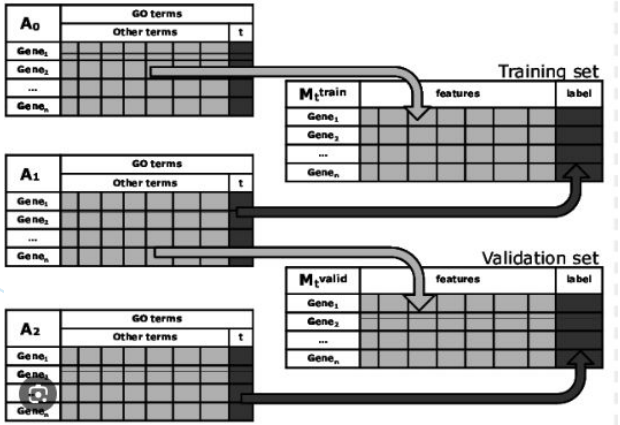
RECORD LESSON

# Agenda

- The importance of Exploratory Data Analysis
- Data Cleaning
- Data Processing and the Concept of ETL
- Data Visualization

# What is EDA?

Exploratory Data Analysis (EDA) is a crucial step in the data analysis process, which involves examining the main characteristics of a dataset, often visually, before making any assumptions or building statistical models. This approach helps in uncovering patterns, spotting anomalies, testing hypotheses, and checking assumptions through summary statistics and graphical representations.

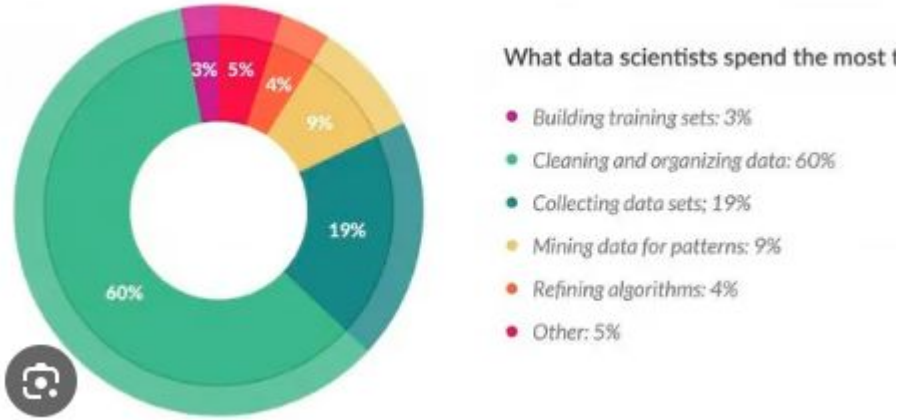



- How many rows?
- What is the avg of “Total”? Does it make sense?
- Is there any data missing in variable “C”?

Welcome

# Data Cleaning

*Assume the data is dirty* - but what does this mean?
You will be faced with datasets of inconsistent data, that you will need to fix - aka, clean - yourselves.



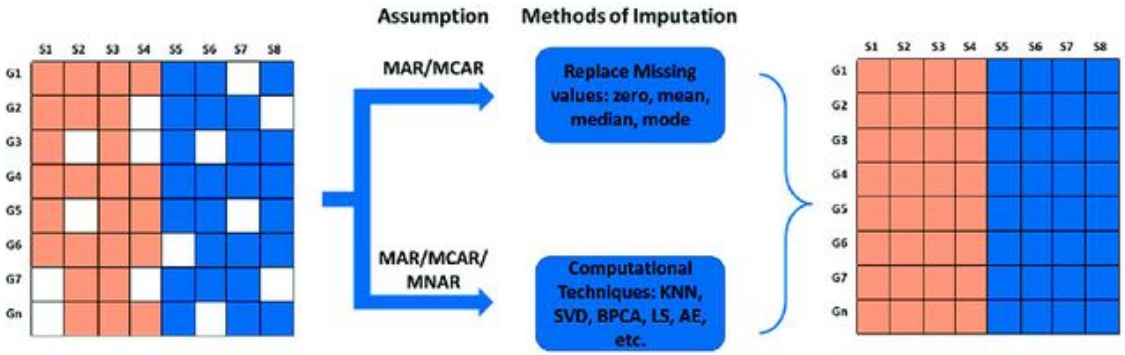

1. San Francisco, CA, USA
2. San Franciso, California, USA (typo)
3. san francisco, california, usa (case issues)
4. SF, California, USA (abbreviation)
5. San Francisco CA USA (missing commas)
6. S. Francisco, California, USA (shortening)
7. San Francisco, Calif., USA (different abbreviation)
8. San Francisco - California, USA (alternative delimiter)
9. san francisco, ca, usa (case and abbreviation issues)
10. San Fran, CA, USA (nickname)
11. SAN FRANCISCO, CALIFORNIA, USA (all caps)
12. San Francisco California, USA (missing comma)
13. San Fransisco, California, USA (typo)
14. Saint Francisco, California, USA (formal/incorrect name)
15. SFO, California, USA (airport code used as city name)
16. San Francisco, Cali, USA (informal abbreviation)
17. San Fran., Cal., USA (abbreviated with periods)
18. San Francisco Calif USA (missing punctuation)
19. San Francisco, Cal., US (country abbreviation inconsistency)
20. San Francsco, California, US (typo and country abbreviation inconsistency)

# Data Injection

*Cleaning data is no just about typos* - often you will also having missing data that is central to your analysis.
“Something” is often better than nothing. But which “something?

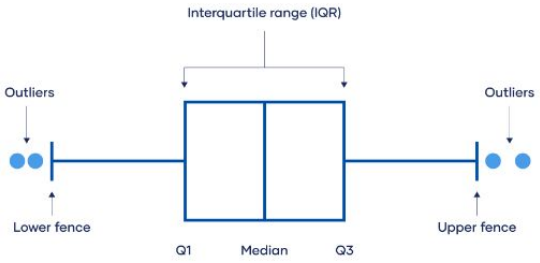
*Common imputation techniques:*

- *Mean*
- *Median*
- *Mode*
- *Zero*
- *and many more*



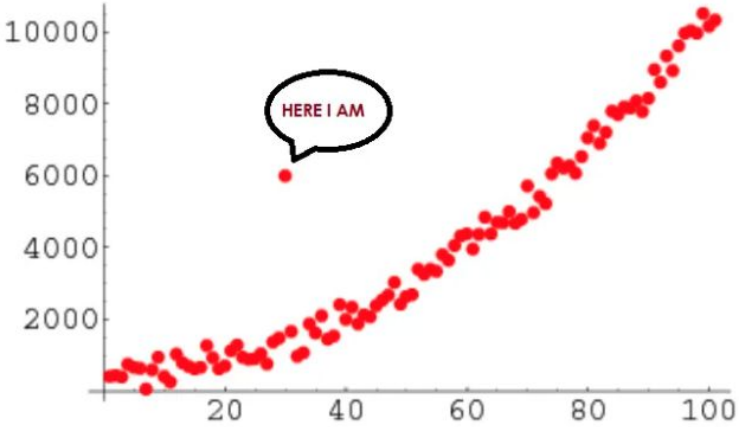
# Outliers

The concept of an outlier is a point that is valid - it exists in our data and in reality - but it is not **representative of our dataset.**



**Dan Price**
@DanPriceSeattle

out of curiosity I ran the numbers and Mark Zuckerberg single-handedly has 2% of all Millennial wealth



## Outliers

What should we do about Outliers?

There is no Golden Rule – Outliers are still valid situations in our reality

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## Outliers

What should we do about Outliers?

Sales of a shop

vs

Wealth of Millenials

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# Data processing - Filtering

Filtering is one of the main parts of EDA - it reflects our choice over the analysis that we want to do and which data we want to include.

**Filtering is done by rules, not by hand**

- Filter on a subset of elements
- Filter on a condition (a > b) or a combination of condictions
- Filter on an issue (remove Nas or invalid rows)

**Filtering is done on the rows**

| | | | | Brand Ratings | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| ID | Age | Gender | Preferred cola | Coca-Cola | Diet Coke | Coke Zero | Pepsi | Diet Pepsi | Pepsi Max |
| 1 | 25 to 29 | Female | Pepsi Max | 2 | 5 | 2 | 3 | 1 | 4 |
| 2 | 45 to 49 | Male | Pepsi Max | 5 | 1 | 5 | 5 | 3 | 4 |
| 3 | 25 to 29 | Female | Diet Coke | 5 | 4 | 2 | 3 | 1 | 1 |
| 4 | 25 to 29 | Female | Coca-Cola | 4 | 2 | 2 | 2 | 2 | 2 |
| 5 | 55 to 64 | Female | Diet Coke | 3 | 4 | 3 | 3 | 4 | 2 |
| 6 | 55 to 64 | Female | Diet Pepsi | 3 | 3 | 3 | 3 | 4 | 4 |
| 7 | 50 to 54 | Female | Coke Zero | 2 | 3 | 5 | 2 | 2 | 2 |
| 8 | 35 to 39 | Female | Coca-Cola | 4 | 2 | 5 | 3 | 2 | 5 |
| 9 | 65 or more | Male | Diet Pepsi | 5 | 5 | 3 | 5 | 5 | 3 |
| 10 | 45 to 49 | Female | Coke Zero | 4 | 4 | 4 | 5 | 5 | 3 |
| 11 | 45 to 49 | Male | Coca-Cola | 4 | 1 | 1 | 4 | 1 | 1 |
| 12 | 55 to 64 | Male | Coca-Cola | 5 | 2 | 2 | 5 | 2 | 2 |
| 13 | 55 to 64 | Male | Coca-Cola | 5 | 2 | 2 | 3 | 2 | 2 |
| 14 | 30 to 34 | Male | Pepsi Max | 3 | 2 | 5 | 3 | 3 | 5 |
| 15 | 65 or more | Female | Diet Pepsi | 2 | 4 | 2 | 5 | 4 | 2 |

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# Data processing - Aggregations

The importance of aggregating data cannot be overstated. Every time technology advances, humanity is able to collect and process a high and more detailed volume of data – but the human brain still needs to understand high level, aggregated data to have a full picture.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| France | TRUE | couplerio-sandb | 2915 | 29 | Hand LLC deal | 22 | USD | 8/31/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 2947 | 30 | Ortiz, Farrell anc | 669 | USD | 8/31/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1604 | 28 | Roob-Nader dea | 622 | USD | 7/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 2995 | 30 | Medhurst-Padbe | 830 | USD | 9/30/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 3093 | 29 | Hintz, Doyle and | 497 | USD | 10/31/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 3112 | 29 | Parker-Schmelei | 75 | USD | 10/31/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 3123 | 29 | Kreiger-Sauer d | 142 | USD | 10/31/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 3142 | 29 | Keeling, Crooks | 51 | USD | 10/31/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 3170 | 29 | Wisozk-Ullrich d | 228 | USD | 10/31/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1628 | 28 | Adams, Kling an | 628 | USD | 7/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 3245 | 29 | Harvey, Wolf anc | 647 | USD | 11/30/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 3283 | 30 | Schinner, Glover | 945 | USD | 11/30/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 3294 | 29 | Mitchell-Purdy d | 867 | USD | 11/30/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 2915 | 30 | Hand LLC deal | 39 | USD | 11/30/2019 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1644 | 28 | Bartoletti-Harris | 287 | USD | 7/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1647 | 28 | Bauch-Casper d | 881 | USD | 8/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1679 | 28 | Stroman-Heaney | 763 | USD | 8/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1698 | 28 | Brakus, Fay and | 340 | USD | 8/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1704 | 30 | Mante and Sons | 439 | USD | 8/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1708 | 28 | Hartmann-Nitzsc | 758 | USD | 8/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1714 | 30 | Adams-Graham | 295 | USD | 8/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1726 | 28 | Hill LLC deal | 148 | USD | 8/31/2018 | 10/18/2019 |
| France | TRUE | couplerio-sandb | 1733 | 28 | Walsh and Sons | 83 | USD | 9/30/2018 | 10/18/2019 |

| Country | Conversion rate | Total revenue |
|---|---|---|
| Australia | 32.01% | $42,851.00 |
| Canada | 28.57% | $38,630.00 |
| Denmark | 28.47% | $34,078.00 |
| France | 27.89% | $39,561.00 |
| Germany | 30.10% | $43,460.00 |
| Netherlands | 28.09% | $45,102.00 |
| Ukraine | 28.04% | $31,025.00 |
| United Kingdom | 33.44% | $52,149.00 |
| United States | 29.50% | $40,088.00 |
| United Arab Emirates | 27.84% | $38,934.00 |

| Total deals | Total revenue | Average deal life time (days) |
|---|---|---|
| 299 | $45,102.00 | 50 |

Welcome

# Data processing – Transformation

During our data flow, transformations will be necessary – and they start in our exploratory analysis.

**Transformations / Formulas:** often we will need to alter what the data looks like

| Date of birth | becomes | Age |
|---|---|---|
| 02/07/1988 | | 35 |

Welcome

# Data processing – Transformation

During our data flow, transformations will be necessary – and they start in our exploratory analysis.

**Transformations / Formulas:** often we will need to alter what the data looks like.

**Statistical Analysis:** normalization, standardization

| Wealth Rage | becomes | Wealth Rage normalized |
|---|---|---|
| 30$-13B$ | | 0-1 |

# Data Visualization

The outcome of all Data Analysis it for a user make an informed decision – that information needs to be shared with the user in some sort of visual matter – often through charts.

The way we visualize that data is, in itself, a science: Data Visualization

Welcome

# Data Visualization

As we study how to visualize data, there are several areas of DataViz to be studied.

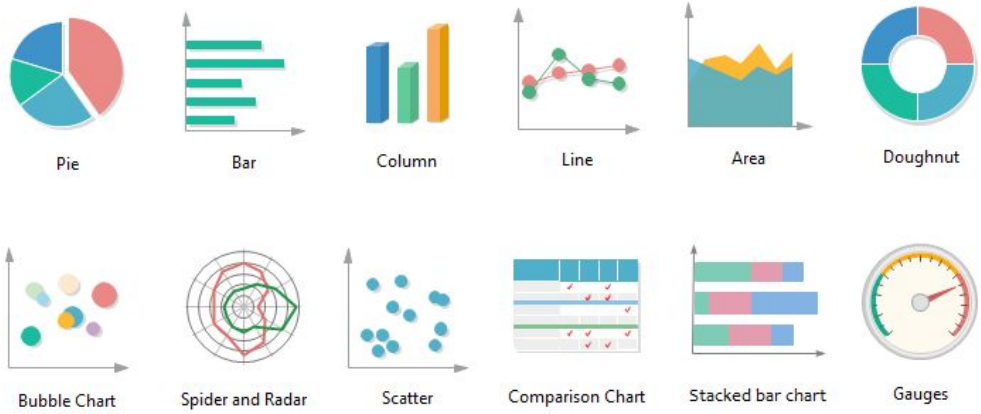
What can be displayed:
DIMENSIONS:
    Data Attributes
    Examples: Geo-Location (City, Country), Categorical Breakdown
    (Usually) strings METRICS:

METRICS
    Measurable values that describe your data
    Examples: Sales, Profit, dedicated KPIs (what are KPIs??)

# Data Visualization

As we study how to visualize data, there are several areas of DataViz to be studied.

The different ways it can be displayed



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# Thank you

Let’s get started!